

Summer Bootcamp – Session 1 Exercises

Linear Algebra

Nathan Sauldubois

August 2026

Instructions

The goal is to become operational with matrix language. Exercises 1–6 are the core paper exercises. Exercises 7–8 are hands-on Python exercises. Exercises 9–11 are decomposition brainteasers and applications. Exercises 12–13 are finance-flavored brainteasers and theoretical questions. Exercises 14–16 connect transition matrices with Markov chains and hitting-time puzzles.

For the hands-on exercises, use only basic NumPy; do not use a built-in PCA or portfolio-optimization function unless explicitly asked to compare.

1 Core Linear Algebra

Exercise 1 – Matrix arithmetic and dimension checks. Let

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix},$$

and

$$x = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}, \quad y = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

1. For each expression, say whether it is defined and give its dimension:

$$AB, \quad BA, \quad AC, \quad CA, \quad Ax, \quad Ay, \quad Bx, \quad By, \quad C^2, \quad A^\top C.$$

2. Compute all defined products among AB , BA , Ax , By , and C^2 .
3. Compute $A^\top$, $B^\top$, $\text{tr}(C)$, $\det(C)$, and C^{-1} .
4. Use your computations to show explicitly that matrix multiplication is not commutative.
5. Interpret Ax as a linear combination of the columns of A .

Exercise 2 – Kernel, image, rank, and geometry. Consider

$$L = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

1. Solve $Lx = 0$ and give a basis of $\ker(L)$.
2. Compute $\text{rank}(L)$ using rank-nullity.
3. Show that every column of L is orthogonal to $(1, 1, 1)^\top$.
4. Deduce that

$$\text{Im}(L) = \{y \in \mathbb{R}^3 : y_1 + y_2 + y_3 = 0\}.$$

5. For each vector below, decide whether $Lx = b$ has a solution:

$$b_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

When a solution exists, find all solutions.

6. Give a geometric interpretation: which direction is killed by L , and on which plane does L act non-trivially?

Exercise 3 – Linear systems and bootstrapping. Part A: a small system. Consider

$$\begin{cases} 2x_1 + 3x_2 = 5, \\ x_1 - 4x_2 = -2. \end{cases}$$

1. Write the system as $Ax = b$.
2. Compute $\det(A)$ and decide whether the solution is unique.
3. Solve the system by substitution.
4. Compute A^{-1} and verify that $x = A^{-1}b$ gives the same answer.

Part B: fixed-income bootstrapping. You observe three annual coupon bonds with face value 1:

Maturity	Coupon	Cash flows	Price
1	3%	1.03 at T_1	1.005
2	4%	0.04 at T_1 , 1.04 at T_2	1.018
3	5%	0.05 at T_1 , 0.05 at T_2 , 1.05 at T_3	1.025

Let $d_k = D(T_k)$ be the discount factor at maturity T_k .

1. Write the pricing equations in matrix form $Cd = p$.
2. Explain why the system is triangular.
3. Solve for d_1, d_2, d_3 using forward substitution.
4. Compute the zero rates

$$z_k = -\frac{1}{T_k} \log d_k, \quad k = 1, 2, 3.$$

5. Explain why this is a linear algebra version of bootstrapping.

Exercise 4 – Covariance matrices and portfolio variance. The following centered return matrix contains four daily observations of two assets:

$$X = \begin{pmatrix} 0.01 & -0.02 \\ 0.02 & 0.01 \\ -0.01 & 0.00 \\ -0.02 & 0.01 \end{pmatrix}.$$

1. Check that each column is centered.
2. Compute

$$\hat{\Sigma} = \frac{1}{n-1} X^\top X, \quad n = 4.$$

3. Verify directly that $\hat{\Sigma}$ is symmetric.
4. For $w = (0.6, 0.4)^\top$, compute $w^\top \hat{\Sigma} w$.
5. Decompose the variance into individual variance terms and covariance terms:

$$w^\top \hat{\Sigma} w = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_{12}.$$

6. Explain why covariance can either increase or reduce portfolio risk.

Exercise 5 – Diagonalization and covariance interpretation. For $\rho \in (-1, 1)$, consider the two-asset correlation matrix

$$C_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$

1. Compute the eigenvalues and eigenvectors of C_ρ .
2. Specialize to $\rho = 0.8$ and write an orthogonal diagonalization $C_\rho = QDQ^\top$.
3. Interpret the two eigenvectors as the common direction and the spread direction.
4. Which eigenvalue becomes small when ρ is close to 1? What does that mean financially?
5. What happens when $\rho = -1$? Is the matrix invertible?

Exercise 6 – PCA by hand. The centered data matrix is

$$X = \begin{pmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{pmatrix}.$$

1. Compute $\hat{\Sigma} = \frac{1}{n-1} X^\top X$.
2. Find its eigenvalues and orthonormal eigenvectors.
3. Identify the first principal component.
4. Project the three observations onto the first principal component.
5. Reconstruct the observations using only the first principal component. Is the reconstruction exact?
6. Explain what information is kept and what information is lost.

2 Hands-On Python

Exercise 7 – Empirical covariance and PCA with NumPy. Use the following return matrix. Rows are days and columns are assets:

$$R = \begin{pmatrix} 0.010 & -0.004 & 0.006 & 0.012 \\ -0.006 & 0.003 & -0.002 & -0.008 \\ 0.014 & 0.001 & 0.010 & 0.015 \\ -0.012 & -0.006 & -0.007 & -0.010 \\ 0.008 & 0.004 & 0.005 & 0.009 \\ 0.003 & -0.002 & 0.001 & 0.004 \\ -0.009 & 0.002 & -0.004 & -0.011 \\ 0.011 & 0.005 & 0.007 & 0.013 \end{pmatrix}.$$

1. Load this matrix into a NumPy array.
2. Compute the empirical mean vector.
3. Center the matrix manually.
4. Compute the empirical covariance matrix using

$$\hat{\Sigma} = \frac{1}{n-1} R_c^\top R_c.$$

5. Compare your result with `np.cov(R, rowvar=False)`.
6. Compute eigenvalues and eigenvectors using `np.linalg.eigh`.
7. Sort eigenvalues in decreasing order and compute the explained variance ratios.
8. Project the centered data onto the first one and then the first two principal components.
9. In two or three sentences, interpret the first principal component as a financial factor.

Suggested skeleton:

```
import numpy as np
```

```

R = np.array([
    [ 0.010, -0.004,  0.006,  0.012],
    [-0.006,  0.003, -0.002, -0.008],
    [ 0.014,  0.001,  0.010,  0.015],
    [-0.012, -0.006, -0.007, -0.010],
    [ 0.008,  0.004,  0.005,  0.009],
    [ 0.003, -0.002,  0.001,  0.004],
    [-0.009,  0.002, -0.004, -0.011],
    [ 0.011,  0.005,  0.007,  0.013],
])

n, d = R.shape
mu = ...
Rc = ...
Sigma = ...
eigvals, eigvecs = ...

```

Exercise 8 – Portfolio risk lab. Consider four assets with expected returns and covariance matrix

$$\mu = \begin{pmatrix} 0.05 \\ 0.07 \\ 0.09 \\ 0.12 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.040 & 0.012 & 0.018 & 0.010 \\ 0.012 & 0.060 & 0.020 & 0.016 \\ 0.018 & 0.020 & 0.090 & 0.030 \\ 0.010 & 0.016 & 0.030 & 0.160 \end{pmatrix}.$$

1. Check numerically that Σ is symmetric.
2. Check numerically that all eigenvalues of Σ are positive.
3. For

$$w_0 = (0.25, 0.25, 0.25, 0.25)^\top,$$

compute expected return, variance, and volatility.

4. Compute the unconstrained minimum-variance portfolio with budget $\mathbf{1}^\top w = 1$:

$$w^* = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}.$$

Use `np.linalg.solve` rather than explicitly forming the inverse.

5. Compare w^* to w_0 in terms of return and volatility.
6. Generate 5000 random long-only portfolios, normalize each one so weights sum to 1, and plot or print the lowest volatility you find.
7. Does the best random portfolio get close to w^* ? Explain why or why not.

3 Decomposition Brainteasers and Applications

Exercise 9 – Predicting a coupled system without iterating. Two quantities evolve according to

$$x_{k+1} = Ax_k, \quad A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}, \quad x_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

1. Compute x_1, x_2, x_3 directly and guess the form of x_k .
2. Diagonalize A and use the decomposition to compute A^k .
3. Deduce a closed formula for x_k and verify your initial guess.
4. Which eigenvalue controls the long-run growth? Explain without carrying out any additional matrix multiplication.
5. Suppose instead that $x_0 = (-1, 1)^\top$. What simplification occurs, and why?

Exercise 10 – Finding hidden risk factors. Two standardized asset returns have correlation matrix

$$C = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}.$$

1. Without expanding a determinant, propose two orthogonal candidate eigenvectors using the symmetry between the assets.
2. Normalize these vectors and compute their eigenvalues.
3. Interpret the two eigenportfolios as a common trade and a spread trade.
4. Compute the variance of each unit eigenportfolio.
5. A trader claims that the equally weighted long-short portfolio is almost riskless. Is that statement literally correct? Quantify your answer.
6. Repeat the reasoning for $C_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ and describe what happens as $\rho \rightarrow 1$ and as $\rho \rightarrow -1$.

Exercise 11 – Manufacturing a target correlation. Let Z_1, Z_2 be independent standard Gaussian variables. You want to construct a Gaussian vector $X = (X_1, X_2)^\top$ with covariance matrix

$$\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}.$$

1. Find the lower-triangular Cholesky factor L such that $\Sigma = LL^\top$.
2. Write $X = LZ$ component by component.
3. Compute $\text{Var}(X_1)$, $\text{Var}(X_2)$, and $\text{Cov}(X_1, X_2)$ directly from your formulas and verify that they match Σ .
4. Compute the correlation between X_1 and X_2 .
5. If $z = (1, -1)^\top$ is one simulated independent shock, compute the corresponding correlated shock Lz .
6. Explain why multiplying by Σ itself, instead of L , would not produce covariance Σ .

4 Finance Brainteasers and Theory

Exercise 12 – Singular covariance and redundant assets. Let R_1 and R_2 be two returns with

$$\text{Var}(R_1) = 1, \quad \text{Var}(R_2) = 1, \quad \text{Cov}(R_1, R_2) = 0.2.$$

Define a third return by $R_3 = R_1 + R_2$.

1. Compute $\text{Cov}(R_1, R_3)$, $\text{Cov}(R_2, R_3)$, and $\text{Var}(R_3)$ from the identity $R_3 = R_1 + R_2$.
2. Use these quantities to write the full covariance matrix Σ of (R_1, R_2, R_3) .
3. Write $R = (R_1, R_2, R_3)^\top$ as $R = MX$, where $X = (R_1, R_2)^\top$, and deduce the factorization $\Sigma = M\Sigma_X M^\top$.
4. Find a nonzero vector w such that $\Sigma w = 0$ and verify the matrix multiplication explicitly.
5. Compute and interpret the corresponding portfolio return $w^\top R$.
6. Explain in three equivalent ways why Σ is singular: using the linear relation among returns, the kernel, and the rank of M .
7. Suppose three securities have payoffs exactly R_1, R_2, R_3 and current prices p_1, p_2, p_3 . Derive the price relation required by no arbitrage.
8. If $p_3 < p_1 + p_2$, give the positions, initial cash flow, and terminal payoff. Repeat the analysis for $p_3 > p_1 + p_2$.

Exercise 13 – Short theoretical questions. Give short but rigorous answers.

1. Prove that for any real matrix X , the matrix $X^\top X$ is symmetric positive semidefinite.
2. Prove that if A is injective, then $\ker(A) = \{0\}$, and prove the converse.
3. Let $A \in \mathbb{R}^{m \times n}$. Explain why $Ax = b$ has a solution if and only if $b \in \text{Im}(A)$.
4. Let A be real symmetric. Prove that eigenvectors associated with distinct eigenvalues are orthogonal.
5. Give an example of a matrix with one repeated eigenvalue that is not diagonalizable.
6. Explain, in one paragraph, why PCA is just diagonalization of a covariance matrix.

5 Markov Chains and Matrix Brainteasers

Throughout this section, probability distributions are column vectors and $P_{ij} = \mathbb{P}(X_{n+1} = i \mid X_n = j)$.

Exercise 14 – Credit migration by matrix powers. Consider

$$P = \begin{pmatrix} 0.9 & 0.4 \\ 0.1 & 0.6 \end{pmatrix}.$$

1. Verify that P is column-stochastic and explain the meaning of each column.
2. Starting from $\mu_0 = (1, 0)^\top$, compute μ_1 and μ_2 .
3. Find the stationary distribution π by solving $P\pi = \pi$ together with $\mathbf{1}^\top \pi = 1$.
4. Find both eigenvalues of P and diagonalize it.
5. Obtain a closed formula for $\mu_n = P^n \mu_0$.
6. How many periods are needed before the Bad-state probability is within 10^{-3} of its stationary value?

Exercise 15 – Waiting time for PPF. A fair coin is tossed repeatedly until the consecutive pattern PPF appears, where P denotes heads and F denotes tails.

1. Define states according to the longest suffix that is also a prefix of PPF.
2. Draw the transition graph and write its transition matrix, including an absorbing completion state.
3. Let E_0, E_P, E_{PP} be the expected remaining waiting times. Condition on the next toss to obtain a linear system.
4. Solve the system and compute the expected number of tosses from the start.
5. Explain why observing another P from state PP does not reset the process.

Exercise 16 – Waiting time for 564. A fair six-sided die is rolled repeatedly until three consecutive rolls are 5, then 6, then 4.

1. Construct states 0, 5, and 56 using the longest useful suffix.
2. Write the transition matrix after adding an absorbing state for 564.
3. Let E_0, E_5, E_{56} be the expected remaining waiting times. Derive the three first-step recurrences.
4. Solve the resulting linear system.
5. Compare the answer with 6^3 and explain the relationship.